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ECOP: Energy-Efficient Computation Offloading with Privacy Awareness for IoT Applications

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***Abstract*—The ever-growing computational demands of consumer electronics in Internet of Things (IoT) applications present a critical challenge, largely attributed to the inherent limitations in the computational capability and energy resources of such devices. Mobile Edge Computing (MEC) has emerged as a promising solution to enhance computational efficiency and support real-time processing. While computation offloading can alleviate device-level processing burdens, it also introduces potential vulnerabilities regarding location data exposure, thereby compromising user privacy. Therefore, to promote the sustainable and efficient operation of electronic devices and enhance the protection of location privacy, this paper proposes an energy-efficient computation offloading scheme with privacy awareness. This scheme achieves adaptive dynamic computation offloading and resource scheduling mechanism for differentiated computational demands by jointly optimizing channel resource allocation, battery state, energy harvesting level, and privacy entropy. The joint optimization scheme not only maintains near-optimal latency and energy efficiency but also significantly enhances the system's privacy entropy. Moreover, this work investigates the impact of RF-enabled Wireless Energy Transfer Access Points (WET-APs) on location privacy leakage. Simulation results demonstrate that ECOP significantly improves computation latency, reduces energy consumption, and strengthens location privacy preservation compared to existing approaches.**

***Index Terms*—Internet of Things, Mobile Edge Computing, Computation offloading, Privacy awareness.**

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I. Introduction

THE widespread adoption of Internet of Things (IoT) technology [1] has triggered an exponential increase in IoT device diversity and catalyzed unprecedented growth in consumer electronics. Modern life now revolves around ubiquitous electronic devices including smartphones, wearable bands, smartwatches, and connected home appliances. Meanwhile, technological advancements have rendered these electronic devices [2] increasingly sophisticated, resulting in three critical challenges: intensive computational requirements, excessive energy consumption, and heightened privacy risks —particularly evident in facial recognition and contactless payment systems. Given the inherent limitations in computational resources and battery capacity, delivering efficient computing services while guaranteeing privacy and security has emerged as a paramount challenge for next-generation consumer electronics [3] [4].

The advent and widespread use of Mobile Edge Computing (MEC) [5] have broadened the potential for IoT applications and offered enhanced support for computation-intensive and energy-demanding consumer electronics. In recent years, IoT applications bolstered by mobile computing have garnered considerable research interest. MEC has emerged as an effective complement to Mobile Cloud Computing (MCC) [6] [7], aiding in faster data processing, shorter latency, lower energy consumption, and reduced bandwidth requirements, which are essential for more responsive and flexible IoT systems. Deploying edge intelligence-based IoT systems is a key strategy for mitigating inefficiencies in electronic devices. Nevertheless, conventional wired charging remains infeasible for body-embedded or wearable electronics. In this context, Energy Harvesting (EH) [8] [9] has emerged as a promising solution to prolong battery life in such devices and enhance Quality of Service (QoS) for users [10].

Integrating energy harvesting allows electronic devices to achieve satisfactory continuous computational performance. However, prior research has primarily concentrated on optimizing energy consumption, computational latency, and the trade-offs between these aspects. Most existing work [11] [12] [13] has concentrated on traditional security concerns transferred from cloud computing to MEC systems, failing to account for the unique privacy vulnerabilities introduced by dynamic offloading policies. Recent analytical studies of offloading behaviors reveal a strong correlation between wireless channel quality and computational offloading volume, where devices exhibit maximal offloading propensity under favorable

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channel conditions [14]. Given the proven relationship between wireless channel gain and transmitter-receiver distance, hackers can implement sophisticated traffic analysis attacks by monitoring task offload characteristics on curious but honest MEC servers. Such inference attacks can reconstruct sensitive channel characteristics, constituting a fundamental breach of privacy that cannot be mitigated by current computational offload architectures [15] [16].

To enable privacy awareness task offloading in energy-constrained environments, significant research has developed optimized computation offloading frameworks incorporating privacy awareness mechanisms. The approach [17] addresses IoT devices' limitations by leveraging MEC for intensive computing tasks and proposes a Lyapunov-based algorithm for real-time task offloading decisions in MEC systems with energy harvesting, enhancing throughput and stability without prior system statistics. However, the amount of energy harvested from environmental energy sources depends significantly on the stability of the energy supply. For instance, solar energy harvesting exhibits strong geographical dependencies and temporal variability, while mechanical energy harvesting offers limited energy density with narrow applicability. These fundamental constraints of green EH technologies [18] create challenges in maintaining reliable power supplies for mission-critical applications, particularly in healthcare monitoring scenarios where energy interruptions may compromise timely medical intervention, potentially exacerbating health outcomes.

The advancement of Radio Frequency Energy Harvesting (RFEH) technology has provided a new possibilities for EH [19] [20]. However, RFEH introduces the risk of location privacy leakage, which is a critical issue addressed in this paper. Electronic devices receive wireless energy from RF-enabled WET-APs, a process whose efficiency depends on wireless channel quality. When multiple MEC servers and WET-APs serve numerous electronic devices across various locations, hackers may infer sensitive location information through repeated computations and iterative analysis. Thus, it is essential for electronic devices to protect their location privacy during the offloading process.

To enable the sustainable operation of electronic devices while bolstering location privacy, this work proposes a privacy-aware computation offloading framework that synergistically integrates Deep Reinforcement Learning (DRL) with RFEH technology. The offloading strategy for each electronic device must consider time-varying information, which is challenging for devices with limited computational and storage capabilities. To cope with these dynamic and uncertain environments, conventional optimization methods are ill-suited as they rely heavily on accurate mathematical models and real-time global state information, which are difficult to obtain in practice. Supervised learning, while fast in inference, depends on large labeled optimal datasets and fails to directly optimize long-term cumulative performance. In contrast, DRL is a model-free paradigm that adaptively learns optimal policies through online interaction, making it inherently suitable for such sequential decision-making problems. Specifically, we adopt the Soft Actor-Critic (SAC) algorithm as the core solver, which incorporates a maximum entropy framework to encourage robust exploration and prevent premature convergence to sub-optimal policies. Compared to alternative DRL variants, SAC offers superior exploration efficiency, hyperparameter robustness, and off-policy sample efficiency, which is critical for online edge computing scenarios. We employ a discrete-action variant of SAC to handle the binary offloading decisions. Notably, unlike prior studies that overlooked the role of WET-APs in privacy leakage, this work systematically investigates how WET-AP deployment affects location privacy when optimizing offloading strategies. This study makes three core contributions to computation offloading systems with privacy-energy co-optimization:

- *Model*: We propose a unified energy-efficient computation offloading with privacy awareness for MEC-assisted IoT systems, which considers various types of tasks with different computational requirements. Specifically, we model and analyze in detail the task computation latency, energy consumption, privacy entropy, and location privacy loss.
- *Optimization Problem*: To counter the threat that attackers could infer users' geographic locations from the publicly accessible network locations they frequently accessed and thereby launch targeted attacks, this paper focused on privacy security issues during computation offloading. To further investigate the system utility, this paper formulated a multi-objective optimization problem that aimed at maximizing privacy entropy while minimizing location privacy loss, computation latency, and energy consumption, and jointly considered multiple factors including local computing, offloading computing, computing resources, and energy harvesting.
- *Algorithm Design*: To tackle the non-convex and NP-hard mixed-integer programming problem, we proposed a SAC-based energy-efficient computation offloading with privacy awareness algorithm (ECOP) to enable dynamic policy selection that perceives environmental states. The proposed scheme offered the advantage of enhancing energy sustainability through adaptive radio frequency energy harvesting scheduling, while simultaneously achieving a higher level of location privacy protection.

The remaining content of this paper is organized as follows: Section II presents a comprehensive review of the related research work. Section III introduces the system models. Section IV proposes an energy-efficient computation offloading with privacy awareness algorithm (ECOP). Section V conducts extensive experiments to verify the proposed method and the Section VI summarizes this research.

II. Related Work

The exponential growth of data generation has exposed critical infrastructure limitations, particularly in mobile architectures where finite processing capabilities increasingly conflict with application requirements for high-throughput task execution and ultra-low service latency [21] [22]. Within this context, such tension has propelled computation offloading to the status of a critical research frontier, given its potential to balance real-time responsiveness with energy conservation.

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Gaurav et al. [23] provided a structured review of contemporary computation resource allocation strategies to inform future research directions. They categorized these strategies according to three criteria: (i) optimization objectives, (ii) employed models and algorithms, and (iii) core technologies involved. Fei et al. [24] introduced a deep reinforcement learning method for task allocation optimization in federated learning, which enhanced processing efficiency while reducing latency for heterogeneous medical data. Quy et al. [25] conducted a comparative analysis of different computing technologies and proposed a unified framework for internet of health things applications. Their findings suggested substantial potential for Fog-IoHT implementations. Hu et al. [26] employed a distributed computing paradigm based on cloud-edge collaboration to optimize edge computing resource utilization.

With the progression of energy harvesting technology, research on mobile computing systems equipped with EH capabilities has attracted significant attention among scholars. Min et al. [27] introduced an offloading algorithm in which IoT devices with EH functionality determine their offloading rates based on battery power levels through reinforcement learning. Zhao et al. [28] developed an edge computing framework that dynamically adjusts video streaming parameters to achieve latency-energy trade-offs through differential privacy mechanisms. They formulated and solved an NP-hard optimization problem through an online learning-based algorithm, demonstrating superior performance compared to existing methods.

The MEC model has attracted an increasing numbers of users due to its distinctive features and rapid service delivery. However, the security of such network architectures is still a significant challenge [29]. He et al. [30] designed a privacy awareness task offloading algorithm aimed at optimizing latency, energy consumption while preserving device privacy by utilizing a constrained Markov decision process. Min et al. [31] introduced a privacy-aware offloading scheme that enables IoT devices to select variable offloading rates for enhanced computing performance and user privacy protection. Notably, the scheme proposed in [31] operates within a green energy-enabled system, which differs from the MEC-assisted IoT systems that require a stable energy supply. Addressing the cyber-spatio-temporal uncertainty in Augmented Reality (AR) applications, the authors in [32] investigated the computational offloading problem within cyber-physical systems, taking into account both user privacy and mobility characteristics. To address location privacy leakage in MEC task offloading, the paper [1] proposed LPA-Offload, which employs differential perturbation for a provable privacy guarantee, supports both single and multi-server scenarios, and adaptively selects perturbation regions to optimize offloading strategies with personalized protection. The paper [33] proposed a DP-based personalized LPTO mechanism for MEC, which jointly optimized perturbation region, privacy budget, and offloading rate via a two-timescale framework and dual-agent DRL, and achieved personalized privacy protection with lower computational cost. To address the overlooked joint optimization of privacy, latency, and energy in IoE offloading, the paper [34] proposed ACTOS, a multi-agent DRL framework with privacy constraints and shared experience replay, validated experimentally. To resist offloading preference inference attacks in VANET, this scheme [35] proposed a privacy-aware MADRL approach that added differential privacy noise to action selection and policy updates, achieving Nash equilibrium with proven privacy guarantees and cost reduction.

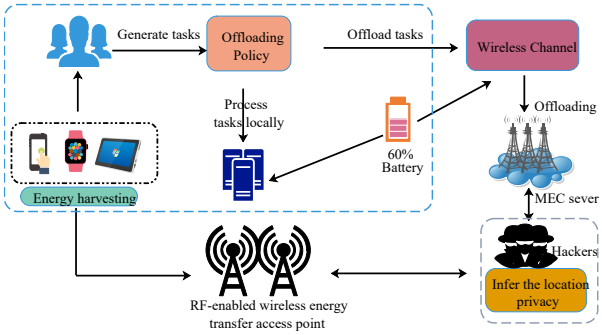


Fig. 1. System modeling of privacy awareness task offloading for MEC-assisted IoT system.

Although substantial work has been conducted on privacy awareness computation offloading, with various models being designed and considered to establish optimization objectives, many existing studies only employed static offloading rates and static privacy parameters to guide the optimization objectives. This approach fails to account for variations between the size of generated tasks and the size of offloaded tasks, resulting in privacy leakage risks under long-term monitoring scenarios. To confront this challenge, we introduce a computation offloading scheme that integrates deep reinforcement learning (DRL) with RFEH technology. This scheme aims to achieve superior system performance and enhanced privacy protection.

III. System Model And Problem Formulation

An MEC-assisted IoT system with energy-efficient and privacy-aware features is considered in this work. As shown in Fig. 1, the MEC-assisted IoT system is composed of a large number of users with electronic devices, MEC servers (ESs), RF-enabled WET-APs and hackers. Each ES can both receive and process task data. These servers provide computational services to electronic devices within their coverage areas, thereby compensating for the devices' limited computing capabilities, reducing processing latency, and extending battery life. To further alleviate the energy supply burden caused by computational demands, all energy-constrained electronic devices are equipped with energy harvesting modules.

In the mobile computing model of this paper, the system time is slotted into ε multiple time slots. During each time slot, every electronic device randomly generates a computation task. We model these time-varying tasks as an independently and identically distributed stochastic process, where each computational task $Z_i(t) = (D_i(t), c_i, T_i^{max})$ can be completed within one time slot. Specifically, $D_i(t)$ (in KB) represents the computation data size and c_i (in CPU cycles/bit) represents the amount of computation resources required for executing each bit of the task. T_i^{max} indicates the maximum tolerable delay of each task. Let the binary variable $a_{ij}(t)$ indicate the

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TABLE I
SCENARIO PARAMETERS

Symbol	Description
$\gamma_{j,i}^{e,t}$	Channel gain between electronic device i and ES j
W	Communication bandwidth between device and ES
δ^2	Gaussian white noise power
$D_i(t)$	Computational task size of electronic device i
c_i	Amount of computation resources required for task
$a_{ij}(t)$	Offloading strategy of computational task i
f_{ij}^{loc}	Local computation resource of device i
$f_{ij}^{edg}(t)$	Allocated computation resource of device i
F_j^{edg}	Ccomputation resource of ES j
$T_{ij}^{loc}(t)$	Local computation delay of computational task i
$T_{ij}^{tra}(t)$	Transmission delay of computational task i
$T_{ij}^{edg}(t)$	Computation delay of ES for computational task i
$E_{ij}^{loc}(t)$	Local energy consumption of computational task i
$E_{ij}^{tra}(t)$	Transmission energy consumption of computational task i
$E_i^{idel}(t)$	Idle energy consumption of computational task i
$p_{j,i}^{e,t}$	Transmit power allocated by ES j
p_i^{idel}	Idle power of device i
V_i^{max}	Maximum battery capacity of electronic device i
$V_i(t)$	Battery level of electronic device i
$v_i(t)$	Harvested energy of electronic device i
ν	Energy conversion efficiency
ρ	Intensity of energy transmission
∂	Path loss factor
G	Combined gain between WET-AP and electronic device
$d_i(t)$	Distance between WET-AP and electronic device i
ε	The total number of time slots

computation offloading decision. If $a_{ij}(t) = 1$, the task $Z_i(t)$ will be all offloaded to the ES j. On the contrary, if $a_{ij}(t) = 0$, the task will be computed locally. The main symbols and parameters throughout this paper are summarized in Tab. I.

Specifically, our framework adopts a centralized control architecture, where the ES acts as the central controller. The ES collects global state information from all connected electronic devices at each time slot. The DRL agent is entirely executed on the ES, including both the training and inference processes. After making the offloading decisions (local computing or offloading computing), the ES sends the binary control commands back to the corresponding devices. This centralized design is practically feasible because electronic devices in our scenario have limited computational capacity and battery life, making it impractical to run DRL models locally. Meanwhile, the ES, typically equipped with powerful computing resources, can efficiently handle the DRL training and inference overhead.

A. Communication Model

To better model the communication connections between electronic devices and ESs while efficiently utilizing channel resources, this work employs Non-Orthogonal Multiple Access (NOMA) technology to establish the communication links between them. The sets of electronic devices, ESs, and subcarriers are denoted as $M = \{1, 2, 3, \cdots, M\}$, $N = \{1, 2, 3, \cdots, N\}$ and $H = \{1, 2, 3, \cdots, H_{sc}\}$, respectively. Under the single-antenna configuration for both ESs and electronic devices, each ES can serve its associated electronic devices over H_{sc} subcarriers, where $M > H_{sc}$ and $N > H_{sc}$. The overall available communication bandwidth is indexed by W, which is uniformly divided into H_{sc} orthogonal multiple access subchannels, available for use by all ESs. The maximum number of electronic devices that can be served by any ES j using its e-th subchannel is defined as N_j^{max} and $N_j^{e,max}$. The maximum transmission power of any ES j is P_j^{max}. To capture the variations in IoT application demands and service deployments at different times, this study divides the system operation time into $T = \{1, 2, \cdots, t\}$ time slots. Specifically, we only consider a time-constrained task processing scheme with limited resources. The computation offloading happens during each time slot. This assumption is justified by our targeted application scenarios, which primarily involve lightweight periodic sensing tasks (e.g., environmental monitoring or wearable health data) that typically exhibit small data sizes and short execution times. Thus, the set of electronic devices served by all subchannels of ES j in time slot t is represented as $M_j^t = \{1, 2, \cdots, M_j^t\}$, where M_j^t denotes the number of electronic devices connected to ES j in time slot t. Considering channel fading, the channel coefficient between any electronic device i served on the e-th subchannel of ES j during time slot t can be defined as:

$$\psi_{j,i}^{e,t} = \eta_{j,i}^{e,t} \zeta_{j,i}^{e,t}, \tag{1}$$

where $\eta_{j,i}^{e,t}$ represents small-scale Rayleigh fading. $\zeta_{j,i}^{e,t}$ determined by path loss and shadowing, belongs to large-scale channel fading.

In time slot t, the superimposed signal transmitted by ES j on its e-th subchannel to the served electronic devices can be expressed as:

$$S_j^{e,t} = \sum\nolimits_{i=1}^{M_j^t} \sqrt{p_{j,i}^{e,t}} \chi_{j,i}^{e,t} s_{j,i}^{e,t}, \tag{2}$$

where $p_{j,i}^{e,t}$ represents the transmission power allocated by ES j in time solt t for serving electronic device i over the e-th subchannel with the signal $s_{j,i}^{e,t}$. $\chi_{j,i}^{e,t}$ is a binary variable reflecting the subchannel allocation decision between ES j and electronic device i in time slot t. $\chi_{j,i}^{e,t} = 1$ indicates that in time slot t, ES j assigns the e-th subchannel to provide service support to electronic device i. Otherwise, $\chi_{j,i}^{e,t} = 0$.

This study categorizes signal interference into two types: inter-cell interference and intra-cell interference. Accordingly, the signal received by electronic device i from the e-th subchannel of the ES j in time slot t can be formulated as:

$$y_{j,i}^{e,t} = \underbrace{\sum_{j' \neq 1, j' \neq j}^{N} \psi_{j',i}^{e,t} x_{j'}^{e,t} P_{j'}^{e,t}}_{\text{inter-cell}} + \underbrace{\sum_{i=1}^{M_j^t} \psi_{j,i}^{e,t} s_{j,i}^{e,t} \sqrt{p_{j,i}^{e,t}} \chi_{j,i}^{e,t}}_{\text{intra-cell}} + \underbrace{\mathrm{I}_{j,i}^{e,t}}_{\text{noise}}, \tag{3}$$

where $\mathrm{I}_{j',i}^{e,t}$ represents the Gaussian white noise experienced by the electronic device i on the e-th subchannel of ES j, which follows $\mathrm{I}_{j',i}^{e,t} \sim N(0, \delta^2)$, where δ^2 is the noise power. $p_{j'}^{e,t}$ is the power allocated by ES j on the e-th subchannel in time slot t. $x_{j'}^{e,t}$ is the transmission signal assigned by ES j on the e-th subchannel in time slot t.

Within the NOMA framework, a hierarchical signal decoding structure is implemented where devices process signals

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in descending order of channel quality. This SIC-based architecture provides dual interference management: (i) intra-cell interference suppression via noise-domain processing and (ii) inter-cell interference elimination through successive decoding iterations. Therefore, the channel gains of electronic device i on the e-th subchannel of ES j in time slot t must satisfy $\gamma^{e,t}_{j,\pi(1)} \leq \gamma^{e,t}_{j,\pi(2)} \leq ... \leq \gamma^{e,t}_{j,\pi(i^t_j)}$. Here, $\pi(i)$ denotes the i-th decoding device, which first decodes the information of the previous $i-1$ devices before decoding its own information. Let d^t_{ji} denote the distance between electronic device i and ES j, $\varkappa$ is the path loss factor. Hence, the Signal-to-Interference-plus-Noise Ratio (SINR) for electronic device i on the e-th subchannel of ES j in time slot t is given by:

$$\gamma^{e,t}_{j,i} = \frac{p^{e,t}_{j,i}\chi^{e,t}_{j,i}|\psi^{e,t}_{j,i}|^2 (d^t_{ji})^{\varkappa}}{\underbrace{\sum_{\pi(i')>\pi(i)} p^{e,t}_{j,i'}\chi^{e,t}_{j,i'}|\psi^{e,t}_{j,i'}|^2}_{\text{intra-cell}} + \underbrace{\sum_{j'\neq 1, j'\neq j}^{N} p^{e,t}_{j'}|\psi^{e,t}_{j',i}|^2}_{\text{inter-cell}} + \delta^2}. \tag{4}$$

Accordingly, the transmission rate for a electronic device i transmitting tasks on the e-th subchannel of the ES j in time slot t can be formulated as:

$$r^{e,t}_{j,i} = \frac{W}{H_{sc}}\log_2(1+\gamma^{e,t}_{j,i}). \tag{5}$$

B. Computation Model

This work does not support task partitioning, and each task is treated as an indivisible whole. For each computation task, the system can actively choose between local computing and offloading to an ES. When neither of these two options is feasible due to insufficient battery power, the task is forced to be abandoned as a last-resort measure. This abandonment is not a decision variable in the optimization; it is an inevitable outcome of resource exhaustion.

Local Computing Model: The electronic device i has a certain computation capacity f^{loc}_i and the computation latency of local computing is expressed as follow:

$$T^{loc}_{ij}(t) = \frac{(1-a_{ij}(t))D_i(t)c_i}{f^{loc}_i}. \tag{6}$$

The corresponding energy consumption of computation task $D_i(t)$ is expressed as:

$$E^{loc}_{ij}(t) = \epsilon_{loc}(f^{loc}_i)^2(1-a_{ij}(t))D_i(t)c_i. \tag{7}$$

The energy consumption per computing cycle is defined as $\epsilon_{loc} = \mathcal{S}f^2$, where $\mathcal{S}$ is the effective switch capacitance that depends on the chip architecture.

Offloading Computing Model: Each electronic device has limited computational resources, making it potentially impossible to execute all tasks locally. Consequently, partial tasks must be offloaded to ES. In this case, the delay of offloading computing includes the upload transmission delay of tasks, the task waiting delay, and computation delay of ES. Execute task by offloading computing model can reduce the heavy workload of electronic devices, but task offloading often requires a relatively large transmission delay. Therefore, we must focus on the trade-off between energy consumption and computation latency.

The upload transmission delay of electronic device i to offload task to ES j is given as:

$$T^{tra}_{ij}(t) = \frac{a_{ij}(t)D_i(t)}{r^{e,t}_{j,i}}. \tag{8}$$

The corresponding transmission energy consumption is represented as follow:

$$E^{tra}_{ij}(t) = p^{e,t}_{j,i}T^{tra}_{ij}(t). \tag{9}$$

Actually, for each task $Z_i(t)$ that arrives at each ES j for processing, it may not always be processed in real time. Whether or not each task can be processed in a timely manner upon arrival at a server depends on the current server execution state. In order to better model the delay consumption of tasks waiting time for processing in the server queue, we model this problem using a queuing model. Let the arrival rate of each task be β and the service rate of the ES be ξ_j. Thus, the service intensity of any ES j can be defined as:

$$\sigma_j = \beta/\xi_j f_j, \tag{10}$$

where f_j reflects the computational power of each ES j.

Define the probability of each ES j having ς tasks as:

$$\sum_{\varsigma=0}^{\infty} P_{j_\varsigma} = 1, P_{j_\varsigma} \geq 0, (\varsigma = 1, 2, \ldots, \kappa), \tag{11}$$

where P_{j_ς} can be formulated as:

$$P_{j_0} = \left[\frac{\beta f_j}{f_j!\,(1-\sigma_j)\,(\xi_A)^{f_A}} + \sum_{\kappa=0}^{f_j-1}\frac{\beta^\kappa}{\kappa!\,(\xi_j)^\kappa}\right]^{\frac{1}{2}}$$
$$P_{j_h} = \begin{cases} \frac{\beta^\kappa \cdot P_{j_0}}{f_j! f_j^{\varsigma-f_j}(\xi_j)^\varsigma}, & \varsigma \geq f_j \\ \frac{\beta^\kappa \cdot P_{j_0}}{\varsigma!(\xi_j)^\varsigma}, & 0 < \varsigma < f_j \end{cases}. \tag{12}$$

Therefore, for each task $Z_i(t)$ the waiting time can be defined as:

$$T^{wit}_{ij}(t) = \frac{(f_j\cdot\sigma_j)^{f_j}\,\sigma_j\cdot P_{j_0}}{f_j!\,(1-\sigma_j)^2\cdot\beta}. \tag{13}$$

The computation delay of ES to execute the offloading task can be calculated as:

$$T^{exe}_{ij}(t) = \frac{a_{ij}(t)D_i(t)c_i}{f^{edg}_{ij}(t)}, \tag{14}$$

where, $f^{edg}_{ij}(t)$ is the computation resources allocated by ES to electronic device i at time slot t.

The corresponding execution energy consumption is represented as follow:

$$E^{exe}_{ij}(t) = \epsilon_{of} f^{edg}_{ij}(t)^{\phi-1}a_{ij}(t)D_i(t)c_i, \tag{15}$$

where ϵ_{of} is the energy coefficient associated with the ES and ϕ is a constant.

We assume that electronic device i stays idle and define the power consumption of the idle state as p^{idle}_i, the corresponding idle energy consumption is calculated as follows:

$$E^{idle}_i(t) = p^{idle}_i T^{exe}_i(t). \tag{16}$$

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Considering the local computing processing and offloading computing processing, the total computation latency and total energy consumption of the system can be expressed as:

$$\begin{aligned} T_{ij}(t) &= \max\{T_{ij}^{loc}(t), (T_{ij}^{tra}(t) + T_{ij}^{wit}(t) + T_{ij}^{exe}(t))\} \\ E_{ij}(t) &= E_{ij}^{loc}(t) + E_{ij}^{tra}(t) + E_{ij}^{exe}(t) + E_{ij}^{idle}(t) \end{aligned} . \tag{17}$$

C. Energy Harvesting Model

Each electronic device is fitted with an energy harvester, which can harvest RF signals from a dedicated RF-enabled WET-Aps. Simultaneously, the signals can be transformed into electronic energy and stored within the battery integrated into electronic devices, ready for utilization when needed. The energy of electronic devices is completely provided by the energy harvester, and the charging and discharging processes of each energy harvester can be performed at the same time. The EH process is modeled as the arrival of continuous energy packets and the harvested energy of different time slots obey the same distribution and statistically independent.

According to the wireless RF energy transmission technology, the energy harvested by electronic device i at time slot t from RE-enabled WET-APs l is expressed as:

$$v_{il}(t) = \nu\rho(d_{il}(t))^{-\partial}G, \tag{18}$$

where, $\nu \in (0, 1)$ represents the energy conversion efficiency, ρ indicates the intensity of energy transmission, ∂ is the path loss factor, G represents the combined gain between WET-AP antenna and electronic device antenna and $d_{il}(t)$ represents the distance between WET-AP and electronic device i, which is modeled as an independent Markov chain with transition probability $P(d_{il}(t+1) = d_{ih}|d(t) = d_{if}) \triangleq p_{fh}$, for $d_{ih}, d_{if} \in \mathcal{G}$.

The battery level of the electronic device i at the beginning of time slot t is denoted as $V_i(t)$. From Eq. 17 and Eq. 18, we can see that the battery state of the device i at the end of time slot t is expressed as:

$$V_i(t+1) = \min\{\max\{V_i(t) + v_{il}(t) - E_{ij}(t), 0\}, V_i^{\max}\}, \tag{19}$$

where, V^{max} represents the maximum battery capacity.

If $V_i(t+1) = 0$, it means that the battery of the electronic device i is exhausted, and the task will be abandoned. The system cost is expressed as:

$$c_i(t) = \omega_1 \mathrm{I}(V_i(t+1) = 0) + \omega_2 T_{ij}(t), \tag{20}$$

where, ω_1 represents the cost when the task is abandoned, $\mathrm{I}(\mu)$ is an indicator function and $\mathrm{I}(\mu) = 1$ when μ is true, otherwise, $\mathrm{I}(\mu) = 0$. ω_2 represents the unit cost of the delay.

IV. Energy-Efficient Computation Offloading with Privacy Awareness

A. Location privacy loss

The propagation of wireless communications is significantly influenced by channel gain $\gamma_{j,i}^{e,t}$, wherein increased distance between users and ES may lead to diminished channel gain, thereby resulting in prolonged latency and heightened energy consumption. Consequently, users typically offload computational tasks to proximate ES to minimize energy consumption and reduce latency. However, this operational paradigm introduces security vulnerabilities, as malicious actors can exploit publicly accessible network locations frequently accessed by users to infer their geographical positioning information, subsequently enabling targeted attacks. To safeguard user location privacy during task offloading processes, we propose a geographical partitioning scheme where the service area is divided into uniform grids, with each grid hosting an ES capable of providing full coverage within its designated spatial domain. Let GI denote the grid zone identifier, which is determined based on whether the geographical location resides within high-risk or sensitive areas (e.g., banks, hospitals, schools) versus ordinary regions. Specifically, $GI = 1$ indicates that the electronic consumer device is currently positioned in a sensitive zone, whereas $GI = 0$ corresponds to regular operational areas.

Since the continuous state transitions of electronic devices' locations exhibit spatiotemporal sequential characteristics-meaning that the current location information depends only on the previous time slot's location. Thus, this paper models the positional data of electronic devices as a Markov chain. Furthermore, as location information constitutes private data known only to the device itself and cannot be directly observed externally (i.e., a hidden state), we further formalize the continuous movement of electronic devices as a hidden markov model. Hackers, leveraging prior knowledge, can obtain the transition probabilities between the initial and hidden states. By analyzing current offloading patterns, they further derive the observation probability matrix that links hidden states (actual locations) to observable states (e.g., connected ESs). Additionally, hackers possess the capability to acquire the set of accessed ES location information, i.e., the observable states-enabling them to infer the device's hidden location states through probabilistic reasoning.

Initial state probability: Let λ_m denote the initial state probability of the electronic device, representing the probability of being at location L_m during the initial time slot $t = 0$. This can be expressed as: $\lambda_m = P(l_t^i = \lambda_m)$.

State transition matrix: Let $s_{nm} = P\{l_{t+1}^i = \lambda_m | l_t^i = \lambda_n\}$ indicate the transition probability of the electronic devices product moving from location L_n to location L_m. The state transition matrix $\mathcal{A} = \{s_{nm}\}$ is derived based on prior knowledge.

Observation probability matrix: Let $b(m, a_{ij}(t)) = P(a_{ij}(t)|l_t^i = \lambda_m)$ represent the the probability that the electronic devices product chooses offloading strategy $a_{ij}(t)$ at time slot t when located at position L_m. Hackers infer based on prior knowledge that products typically select a set of ESs with the minimal latency for task processing. Since ES computational capabilities are heterogeneous, sets with more ES candidates are more likely to be chosen for task offloading. This can be formalized as: $b\left(m, a_{ij}(t)\right) \propto j$.

Location attack probability: The success rate of hacker attacks is closely tied to the task offloading patterns, as demonstrated in the preceding analysis. Consequently, this paper further examines the attack probability from two perspectives:

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local computing mode and offloading computing mode.

1) Local computing mode: Local computation implies no access to ES, so hackers cannot infer the vehicle's location through ES positions, resulting in an observation probability of 0. Additionally, the vehicle's positional changes follow the dynamics of a hidden markov model. Therefore, hackers can infer the success probability $P_{loc}(t, \lambda_m)$ of location inference attack at time slot t using the state transition probability s_{nm} and observation probability $P(t-1, \lambda_n)$ at time slot $t-1$, formalized as:

$$P_{loc}(t, \lambda_m) = \sum_{n=1}^{Y} P(t-1, \lambda_n) a_{ij}(t), \tag{21}$$

where Y is the total number of grids.

2) Offloading computing mode: When the electronic devices product performs offloading computation at time slot t, its transition from location λ_n to λ_m results in corresponding changes in the observation state offloading strategy $a_{ij}(t)$. Therefore, by combining the task offloading state at time slot $t-1$, along with matrices λ_m and $a_{ij}(t)$, the hacker can derive the attack success probability. Hackers leverage prior knowledge of the sequence of offloading strategies $\{a_{ij}(1), \cdots, a_{ij}(t)\}$ prior to time slot t (including time slot t), combined with the location information l_{t-1}^{i} of the consumer electronic product at time slot $t-1$, to successfully infer the probability $P_{of}(t, \lambda_m)$ that the device is located at position λ_m. Assuming the probability at time slot $t-1$ can be expressed as:

$$P_{of}(t-1, \lambda_n) = P\left(a_{ij}(1), \cdots, a_{ij}(t-1), l_{t-1}^{i} = \lambda_n\right). \tag{22}$$

Based on the forward algorithm, the hacker's attack success probability at time slot t can be derived as Eq. 23:

$$\begin{aligned} P_{of}(t, \lambda_m) &= P\left(a_{ij}(1), \cdots, a_{ij}(t), l_{t-1}^{i}, l_{t}^{i} = \lambda_m\right) \\ &= \left[\sum_{n=1}^{Y} P(t-1, \lambda_m) a_{ij}(t)\right] b(m, a_{ij}(t)) \end{aligned}. \tag{23}$$

Intuitively speaking, the leakage of location privacy in electronic devices is directly related to the number of ES they connect to. The more ESs are accessed, the higher the risk of hacker attacks. Consequently, this paper proposes the following theorem.

Theorem 1. *The total location privacy loss ($\mathcal{L}(t)$) of electronic devices during a specific time slot t is directly proportional to the number of ESs accessed; that is, $\mathcal{L}(t)$ incrementally accumulates as the number of connected ESs increases:*

$$\mathcal{L}(t)_{j+1} \geqq \mathcal{L}(t)_{j}. \tag{24}$$

Proof. At time slot t, electronic devices products have two modes for task processing: local computation and offloading computation. However, in the local computation mode, the device does not access any ESs, and thus the location privacy loss in this mode is independent of the number of accessed ES. Therefore, we only need to establish the relationship between location privacy and the number of accessed ES in the offloading mode.

$$\begin{aligned} &\because \begin{cases} \mathcal{L}(t) \propto P_{of}(t, \lambda_m) \\ P_{of}(t, \lambda_m) \propto b\left(m, a_{ij}(t)\right) \end{cases} . \\ &\Rightarrow \mathcal{L}(t) \propto b\left(m, a_{ij}(t)\right) \end{aligned} \tag{25}$$

Since $b\left(m, a_{ij}(t)\right) \propto j$ is closely related to the number of accessed ES, it follows that $b\left(m, a_{ij}(t)\right)_{n+1} > b\left(m, a_{ij}(t)\right)_{j}$ can be derived, which allows us to prove $\mathcal{L}(t)_{j+1} \geqq \mathcal{L}(t)_{j}$. Therefore, theorem. 1 is established. □

Through the above calculation and derivation, it can be concluded that the calculation formula for the location privacy loss incurred by the consumer electronic device-completed task set in time slot t is as follows: $\mathcal{L}(t) = P_{loc}(t, \lambda_m)$ (local computing) or $\mathcal{L}(t) = P_{of}(t, \lambda_m)$ (offloading computing).

B. Privacy Entropy

It was discovered during this investigation that as the distance between the consumer electronic device and the ES changes, so does the wireless channel power gain. If multiple ESs and multiple WET-APs provide services for multiple electronic devices, the location privacy of user can be inferred by these attacked servers and WET-APs. Therefore, it is necessary to fully consider the possibility of the location privacy leakage due to the offloading policy.

In the privacy model of this paper, if the privacy entropy is greater the confusion degree of the task will be greater and the location privacy will be better protected. Privacy entropy is quantified as the following formula:

$$\begin{aligned} p_i(t) = &\omega_3 \cdot |D_i(t) - D_{ij}^{of}(t)| \cdot \mathrm{I}(\gamma_{j,i}^{e,t} > \varrho) \cdot \mathrm{I}(v_{il}(t) > \widetilde{v}) \\ &+ \omega_4 \cdot |D_i(t) - D_{ij}^{of}(t)| \cdot \mathrm{I}(\gamma_{j,i}^{e,t} > \varrho) \cdot \mathrm{I}(v_{il}(t) > \widetilde{v}) \\ &+ \omega_5 \cdot D_{ij}^{of}(t) \cdot \mathrm{I}(\gamma_{j,i}^{e,t} > \varrho) \cdot \mathrm{I}(v_{il}(t) > \widetilde{v}) \\ &+ \omega_6 \cdot D_{ij}^{of}(t) \cdot \mathrm{I}(\gamma_{j,i}^{e,t} > \varrho) \cdot \mathrm{I}(v_{il}(t) > \widetilde{v}) \end{aligned}, \tag{26}$$

where, $D_{ij}^{of}(t) = a_{ij}(t) D_i(t)$, ϱ represents the channel quality threshold. When $\gamma_{j,i}^{e,t} > \varrho$, the wireless channel is considered to be good and all tasks are typically offloaded from the electronic device to the ES for processing. On the contrary, when $\gamma_{j,i}^{e,t} < \varrho$, the electronic device tends to compute all tasks locally. $\widetilde{v}$ represents the harvested energy threshold. When $v_{il}(t) > \widetilde{v}$, the channel condition between the electronic device and the WET-AP is considered to be good and the harvested energy is sufficient. Notably, location privacy of electronic devices can be better protected by adjusting the task offloading rate based on channel conditions. Specifically, a lower offloading rate is preferred in good channel environments, while a higher rate is beneficial when channels are unstable. The greater the offloading frequency deviates from the original offloading frequency, the greater the privacy entropy will be and the lower the risk of locking onto the target user will be.

When $\gamma_{j,i}^{e,t} > \varrho$ and $v_{il}(t) > \widetilde{v}$ the privacy entropy is expressed as the first term of Eq. 26. When $\gamma_{j,i}^{e,t} > \varrho$ and $v_{il}(t) < \widetilde{v}$ the privacy entropy is expressed as the second term of Eq. 26. Both of them represent the difference in data

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size between the tasks generated and the tasks offloaded. ω_3 and ω_4 are weighting factors. When the harvested energy is insufficient and the wireless channel is good, electronic device tend to offload all tasks to improve system utility in the model that does not consider privacy. When the harvested energy is insufficient, more computation tasks should be computed locally to better protect the location privacy in our model. Therefore, we set $\omega_3 < \omega_4$.

When $\gamma_{j,i}^{e,t} < \varrho$ and $v_{il}(t) > \widetilde{v}$ the privacy entropy is expressed as the third term of Eq. 26. When $\gamma_{j,i}^{e,t} < \varrho$ and $v_{il}(t) < \widetilde{v}$ the privacy entropy is expressed as the fourth term of Eq. 26. Both of them represent the data size of the tasks offloaded. ω_5 and ω_6 are weighting factors. When the harvested energy is sufficient and the wireless channel is poor in order to better protect the location privacy more computation tasks should be offloaded. Therefore, we set $\omega_5 > \omega_6$.

From Eq. 20 and Eq. 26, we can see that the utility of the electronic device i at time slot t can be expressed as:

$$U_i(t) = P_i(t) - \omega_1 \mathrm{I}(\mathrm{V_i(t+1)} = 0) - \omega_2 \mathrm{T_{ij}(t)} - \mathrm{GI} \cdot \vartheta \mathcal{L}(\mathrm{t}). \tag{27}$$

where ϑ is a weighting factor representing the privacy importance of sensitive locations.

C. Objective Function

In the energy-efficient and privacy awareness offloading problem considered of this paper, the goal of the MEC-assisted IoT system is to dynamically optimize the offloading policy of electronic devices to maximize the long-term system utility while considering location privacy by analyzing the time-varying harvested energy, time-varying computation tasks and time-varying wireless channel status. To address the offloading policy in the MEC-assisted IoT system, an optimization problem is constructed and expressed in the following form:

$$\begin{aligned}
&\mathcal{P}: \max_{A(t)} \lim_{T\to\infty} \sum_{t=1}^{T}\sum_{j=1}^{N}\sum_{i=1}^{M} U_i(t) \\
s.t. &C1: 0 \le V_i(t) \le V_i^{\max}, \forall i \in M \\
&C2: 0 \le \sum\nolimits_{i=1}^{M}\sum\nolimits_{j=1}^{N} f_{ij}^{edg}(t) \le F_i^{edg}, \forall t \in T \\
&C3: a_{ij}(t) \in \{0,1\}, \forall i \in M, \forall j \in N \\
&C4: 0 \le \sum\nolimits_{i=1}^{M}\sum\nolimits_{j=1}^{N} T_{ij}^{total}(t) \le T_i^{max}, \forall t \in T \\
&C5: \sum\nolimits_{j=1}^{N}\sum\nolimits_{e=1}^{H} \chi_{j,i}^{e,t} \le 1, \forall i, \forall j, \forall e, \forall t \\
&C6: \sum\nolimits_{j=1}^{N} \chi_{j,i}^{e,t} \le N_j^{e,max}, \forall i, \forall j, \forall e, \forall t \\
&C7: \gamma_{j,i}^{e,t} \le \gamma_{j,i'}^{e,t}, \pi(i) \le \pi(i'), \forall (i,i'), \forall j, \forall e, \forall t \\
&C8: \sum_{e=1}^{H}\sum_{i=1}^{M} p_{j,i}^{e,t}\chi_{j,i}^{e,t} \le P_j^{max}, \forall j \in N, \forall e \in H \\
&C9: \sum_{i=1}^{M}\sum_{t=1}^{T} a_{ij}(t) = 1, \forall j \in N, \forall e \in H
\end{aligned} \tag{28}$$

where, C1 represents that the battery level of electronic devices cannot exceed the maximum value V_i^{max}, C2 represents that the sum of computation resource allocated by ES to electronic devices cannot exceed the entire resource F_i^{edg} and the constraint C3 represents the offloading policy of electronic device. C4 reflects that for each task, the latency to complete that task must not exceed its own maximum latency constraint. C5 and C6 reflect that, in any time slot, any electronic device can only connect to one ES and can only access one subchannel of that ES. C7 indicates that SIC technology can achieve signal decoding through specific techniques, while also eliminating intra-cell interference to avoid interference. C8 indicates that the transmission power allocated by the ES to electronic devices must not exceed the ES's maximum transmission power. C9 shows that each electronic devices can be communicatively connected to only one ES. In order to solve the problem $\mathcal{P}$, the system needs to find an offloading strategy in each time slot t, which is mathematically a mapping from the state space $S(t)$ to the action space $A(t)$. Due to the randomness and suddenness of the system parameters, it is impractical to calculate the optimal solution of this problem offline. Therefore, we propose an energy-efficient computation offloading with privacy awareness algorithm (ECOP) to tackle the collaborative optimization of task offloading and privacy protection in MEC-assisted IoT system.

D. Energy-Efficient Computation Offloading with Privacy Awareness

The optimization problem $\mathcal{P}$ is recognized as a mixed integer programming problem, and it is characterized by non-convex[1] and NP-hard problem[2]. Since classical optimization and decomposition methods exhibit inherent limitations in dynamic MEC scenarios: (i) they heavily rely on accurate analytical system models, while our system features highly stochastic wireless channels and time-varying task arrivals; (ii) these algorithms need to be re-executed from scratch once the environment changes, resulting in prohibitive online latency; (iii) they fail to exploit historical environmental data to optimize future decisions. In contrast, DRL adopts a model-free learning paradigm, which transfers heavy computational overhead to the offline training stage and rapidly outputs near-optimal offloading policies within milliseconds during online deployment. Specifically, it can be solved by finding the optimal value of the offloading decision vector $A(t)$. However, it needs to be emphasized that the offloading scheduling strategy of the MEC-assisted IoT system equipped with energy harvester and considering location privacy is much more complicated than the traditional battery-driven system.

This study formulates the dynamic task offloading process as a Markov Decision Process (MDP), aiming to determine the optimal offloading strategy $A(t)$. The ECOP algorithm leverages the current system state to solve the problem at each time slot, enabling the derivation of an approximate optimal

[1]Due to the presence of binary offloading decisions coupled with continuous resource allocation, the feasible solution set is non-convex.

[2]Since the coefficients of problem $\mathcal{P}$ are not totally unimodular matrix (the determinants of any sub-matrix are: 1, 0, 1), thus it is NP-hard.

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solution. The corresponding definitions of the MDP elements are outlined below:

- *state*: At time slot t, the state of the electronic device i is represented as $s_i(t) = \{V_i(t), v_{il}(t), Z_i(t)\}$. It consists of three parts, including the battery level $V_i(t)$, the amounts of harvested energy $v_{il}(t)$ and the generated task $Z_i(t)$. For the entire system, the state space can be expressed as $S_t = \{s_1(t), s_2(t), \ldots, s_M(t)\}$.
- *action*: At time slot t, the subchannel allocation decision between ES j and electronic device i can be indexed as $\chi_{j,i}^{e,t} = \{0, 1\}$. At time slot t, the action of electronic device i is represented as $a_{ij}(t)$ with $a_{ij}(t) \in \{0, 1\}$. It represents the offloading decision that $1 - a_i j(t)$ part of tasks will be computed by local computing and $a_i(t)$ part of tasks will be computed by offloading computing. For the entire system, the action space can be expressed as $A_t = \{\chi_{j,i}^{e,t} = \{\chi_{1,1}^{e,t}, \chi_{2,2}^{e,t}, \cdots, \chi_{N,M}^{e,t}\}, a_{ij}(t) = \{a_{11}(t), a_{22}(t), \cdots, a_{MN}(t)\}\}$.
- *reward*: The reward is the agent's assessment of the benefits obtained in previous action. For each step, the agent will receive a reward $R(S(t), A(t))$ in a certain state $S(t)$ after performing every possible action $A(t)$ at time slot t. Generally speaking, the reward function is related to the objective function. Therefore, the reward function is given by:

$$R_t = \sum_{i=1}^{M} U_i(t). \tag{29}$$

In this paper, we propose our ECOP based on the SAC algorithm. Specifically, problem $\mathcal{P}$ inherently involves a hybrid action space—discrete offloading decisions and continuous resource allocation. The SAC is an off-policy algorithm natively designed for continuous action spaces, whose central idea is to learn a policy through a maximum entropy objective function to achieve better exploration and robustness. SAC incorporates a temperature-parameterized entropy term in its objective. SAC's intrinsic exploration mechanism encourages thorough exploration of the Pareto frontier, preventing premature convergence to myopic policies. The policy network in SAC is modeled as a parameterized Gaussian distribution, used to output the standard deviation and mean of actions. Additionally, SAC maintains two Q-value functions to estimate the value of state-action pairs, along with two target value functions to generate target Q-values. The policy's learning objective seeks to maximize both the cumulative expected reward and the entropy of each action output. Thus, the optimal policy π^* can be defined as:

$$\pi^*(s) = arg\max_{\pi} \sum_{t} E_{(S_t, A_t)\sim\rho}[R(S_t, A_t) + \alpha H(\pi(\cdot|S_t))], \tag{30}$$

where, α is a parameter that quantifies the importance of entropy and controls the stochasticity of the optimal policy. $R(S_t, A_t)$ denotes the reward at time step t. The term $H(\pi(\cdot|S_t)) = E_{A_t\sim\pi(\cdot|S_t)}[-\log(\pi(A_t|S_t))]$ represents the entropy of the policy π given the state S_t.

To mitigate the overestimation of Q-value, this section designs the SAC algorithm to include a policy network and two Q-value function networks. Additionally, to maintain training stability, the algorithm uses target Q-value function networks that correspond one-to-one with the Q-value function networks during the update iterations. Therefore, the SAC algorithm designed in this section consists of: a policy network $\pi_\varphi(A_t|S_t)$, two Q-value function networks $(Q_{\theta_1}(S_t, A_t), Q_{\theta_2}(S_t, A_t))$, and two target Q-value function networks $(Q_{\bar{\theta_1}}(S_t, A_t), Q_{\bar{\theta_2}}(S_t, A_t))$. For each Q-value function network $Q_{\theta_i}(S_t, A_t)$, the parameters θ_i are optimized with the minimization of the following equation:

$$\begin{aligned} J_Q(\theta_i) &= E_{(S_t, A_t, R_t, S_{t+1})\sim D}[\frac{1}{2}(Q_{\bar{\theta_i}}(S_t, A_t) \\ &\quad - (R_t + \gamma V_{Q_{\bar{\theta_i}}}(S_{t+1})))^2] \\ V_{\theta_i^:}(S_{t+1}) &= E_{A_{t+1}\sim\pi_\theta(\cdot|S_{t+1})}[\min_{i=1,2} Q_{\bar{\theta_i}}(S_{t+1}, A_{t+1}) \\ &\quad - \alpha log(\pi(A_{t+1}|S_{t+1}))] \end{aligned}. \tag{31}$$

For the target Q-value function networks $(Q_{\bar{\theta_i}}(S_t, A_t))$, an exponential moving average method is used to stably update the target Q-values, as follows:

$$\bar{\theta_i} = \tau\theta_i + (1 - \tau)\bar{\theta_i}, \tag{32}$$

where, τ is the update factor that regulates the update rate of the target network.

Correspondingly, the policy network $\pi_\varphi(A_t|S_t)$ can be optimized by minimizing Eq. 33:

$$J_\pi(\varphi) = E_{S_t\sim D}[E_{A_t\sim\pi_\varphi}[\alpha log(\pi_\varphi(A_t|S_t)) - \min_{i=1,2} Q_{\theta_i(S_t, A_t)}]]. \tag{33}$$

Considering the dynamic nature of reward values during network training and to ensure stability throughout the process, this section defines α as a variable, automatically adjustable factor. At the early stages of training, a larger α value is set to enhance exploration. As the network stabilizes, α is reduced to ensure a steady state. Initially, α is assigned a random value. When the target entropy is H_0, the α loss is defined as:

$$J(\alpha) = E_{A_t\sim\pi_t}[-\alpha log\pi_t(A_t|S_t) - \alpha H_0]. \tag{34}$$

Algorithm. 1 and Fig. 2 present the detailed training process of the ECOP algorithm. The algorithm's inputs include the system state (ES attributes, electronic device properties, network bandwidth, task size, etc.) and network parameters (policy network $\pi_\varphi(A_t|S_t)$, Q-value function networks $(Q_{\theta_1}(S_t, A_t), Q_{\theta_2}(S_t, A_t))$, and target Q-value function networks $(Q_{\bar{\theta_1}}(S_t, A_t), Q_{\bar{\theta_2}}(S_t, A_t)))$, experience buffer D, parameter α, and others. The algorithm outputs the optimal Q-value function parameters $(\bar{\theta_1}, \bar{\theta_2})$ and the optimal policy network parameters φ^*. Lines 1–4 represent initialization steps. Lines 5–19 describe the training of the network over E episodes. Line 6 initializes the system environment. Lines 7–11 indicate that actions and the environment interact to obtain reward values and the next environment state, with the resulting information stored in the experience buffer D. Lines 12–18 outline the training process, where, in each training iteration, multiple sets of data are sampled from the experience buffer to train the policy network $\pi_\varphi(A_t|S_t)$, the Q-value

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function networks $(Q_{\theta_1}(S_t, A_t), Q_{\theta_2}(S_t, A_t))$, the target Q-value function networks $(Q_{\theta_1'}(S_t, A_t), Q_{\theta_2'}(S_t, A_t))$, and the parameter α. After all training episodes are completed, the optimal Q-value function parameters (θ_1^*, θ_2^*) and the optimal policy network parameters φ^* are returned.

Algorithm 1 ECOP Algorithm

Require: Collection of electronic devices $M = \{1, 2, 3, \ldots, M\}$, collection of ESs $N = \{1, 2, 3, \ldots, N\}$, collection of subchannels $H = \{1, 2, 3, \ldots, H_{sc}\}$, $Z_i(t) = (D_i(t), c_i, T_i^{max})$, $V_i(t)$, $v_{il}(t)$, policy network $\pi_\varphi(A_t|S_t)$, Q-value function networks $(Q_{\theta_1}(S_t, A_t), Q_{\theta_2}(S_t, A_t))$, target Q-value function networks $(Q_{\bar{\theta_1}}(S_t, A_t), Q_{\bar{\theta_2}}(S_t, A_t))$, parameter α, the experience buffer D, communication bandwidth W, maximum battery capacity of electronic device i is V_i^{max} and entire resource of ES j is F_j^{edg}

Ensure: Optimal Q-value function parameters (θ_1^*, θ_2^*) and the optimal policy network parameters φ^*

Initialize policy network parameter: φ, Q-value function network parameters: θ_1^* and θ_2^*
$\bar{\theta_1} \leftarrow \theta_1$, $\bar{\theta_2} \leftarrow \theta_2$ //copy Q-value function network parameters to the target Q-value function network
Initialize the experience buffer D
Initialize the parameter α
for $episode = 1, \cdots, E$ **do**
  Initial state S_0
  for $k = 1$ to K **do**
    $A_t \sim \pi_\varphi(A_t|S_t)$ //sample action A_t from the policy π_φ
    Execute action A_t, interact with the environment to obtain reward R_t, and update the environment state to S_{t+1}
    $D \leftarrow D \bigcup \{(S_t, A_t, R_t, S_{t+1})\}$ // store the new data in the experience buffer
  end for
  for $t = 1$ to T **do**
    Sample N batches of data from D.
    $\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i} J_Q(\theta_i), i \in \{1, 2\}$ //update the Q-value function network
    $\varphi \leftarrow \varphi - \lambda_\varphi \hat{\nabla}_\varphi J_\pi(\varphi)$ //update the policy network
    $\alpha \leftarrow \alpha - \lambda \hat{\nabla}_\alpha J(\alpha)$ //update the parameter α
    $\bar{\theta_i} \leftarrow \tau\theta_i + (1-\tau)\bar{\theta_i}, i \in \{1, 2\}$ //update the target Q-value function network
  end for
end for
return $\varphi^* \leftarrow \varphi$, $\theta_1^* \leftarrow \theta_1$, $\theta_2^* \leftarrow \theta_2$

E. Theoretical Analysis of the Computational Complexity

The computational complexity of the ECOP algorithm can be divided into two main components: the training phase and the execution phase. The training computational complexity represents the computational resources required to optimize the policy and Q-value functions, whereas the execution computational complexity denotes the resources required to generate actions using the learned policy for real-time decision making.

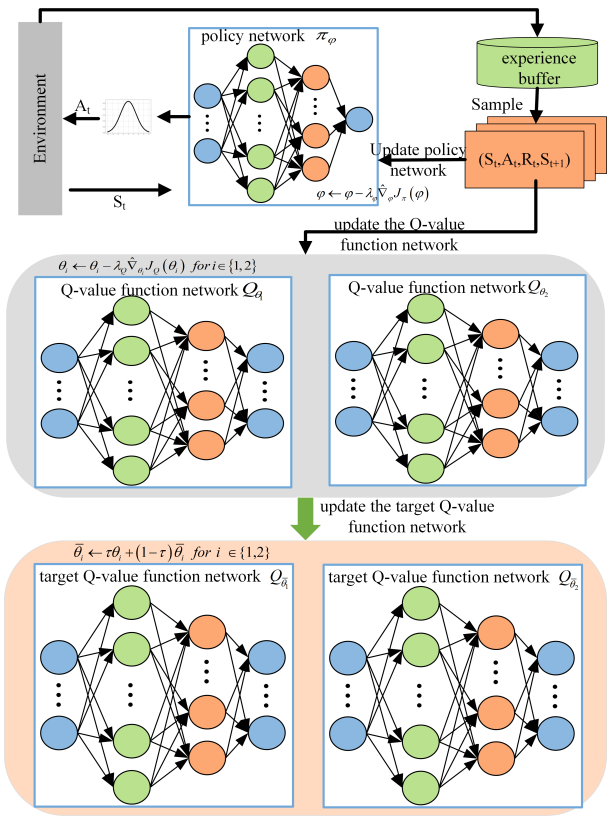


Fig. 2. The workflow of ECOP algorithm.

The policy network is modeled as a parameterized Gaussian distribution that outputs the mean and standard deviation of the action distribution. The two Q-value function networks are used to estimate the value of state-action pairs, while the two target Q-value function networks are used to generate the target Q-values. The policy network and Q-value function networks are alternately optimized to progressively improve policy performance. For the training phase, let the number of layers in the policy network be l_π, and let the number of neurons in the $i-th$ layer be π_i. Similarly, let the number of layers in each Q-value function network be l_Q, and let the number of neurons in the $i-th$ layer be Q_i. The computational complexity of the policy network is $O\left(\sum_{i=1}^{l_\pi - 1} \pi_i \times \pi_{i+1}\right)$. The two Q-value function networks are updated simultaneously. Thus, their computational complexity is $O\left(2K \times T_{update} \times \sum_{i=1}^{l_Q - 1} Q_i \times Q_{i+1}\right)$, where $K \times T_{update}$ denotes the number of parameter-update operations during each training process. In addition, the two target Q-value function networks are used to compute the target Q-values, resulting in a forward-propagation complexity of $O\left(2K \times T_{update} \times \sum_{i=1}^{l_Q - 1} Q_i \times Q_{i+1}\right)$. Let the number of training episodes be EP, the number of steps in each episode be T, and the replay buffer size be RB_{size}. The number of training operations can be approximated as $EP \times T / RB_{size}$. Therefore, the overall computational complexity of the training phase can be expressed as

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$O\left(\frac{EP\times T\times K\times T_{update}}{RB_{size}}\left[\sum_{i=1}^{l_\pi-1}\pi_i\times\pi_{i+1}+4\sum_{i=1}^{l_Q-1}Q_i\times Q_{i+1}\right]\right)$.

For the execution phase, the Q-value function networks and target Q-value function networks are no longer required. Given the current state, the learned policy network only needs to perform one forward propagation to output the mean and standard deviation of the Gaussian action distribution and generate the corresponding action. Hence, the computational complexity of the execution phase is $O\left(\sum_{i=1}^{l_\pi-1}\pi_i\times\pi_{i+1}\right)$. Therefore, although the training phase of the ECOP algorithm has a relatively high computational cost due to the use of two Q-value function networks and two target Q-value function networks, these additional computations are primarily performed offline during training. During online execution, only the policy network is required, resulting in a relatively low computational complexity and enabling ECOP to meet the requirements of real-time decision making.

V. Performance Evaluation

This section focuses on evaluating the performance of the proposed ECOP algorithm. Therefore, to benchmark the proposed solution, we have compared with three competing algorithms, which are described as follows: a) Naive: This algorithm is just a simplification of the energy-efficient computation offloading with privacy awareness algorithm (ECOP) we proposed. b) CMDP [30]: Built upon the CMDP framework, this algorithm treats privacy preservation as a constraint, not an objective. It actively injects controlled obfuscation behaviors to mislead the MEC server, achieving the required long-term privacy level while optimizing latency and energy consumption. c) RL [36]: This algorithm is to achieve joint latency-privacy optimization for multi-user wireless-powered MEC by disguising offloading decision—specifically, by sending a mix of redundant and real tasks to the MEC server, thereby protecting both location and usage-pattern privacy.

A. Simulation Settings

In the simulation, each time slot lasts for 100 ms and the battery capacity of each electronic device is V_i^{max} uniformly distributed in $[10, 20]$ mJ. The total number of time slots ε is set to 6. The number of electronic devices, ESs and RF-enabled WET-APs are set to 150, 10 and 10. The number of subchannel is 38. Each subcarrier has a bandwidth of 20 MHz. Set the cell radius to 1000 m. In each time slot, we set one sub-channel can be jointly occupied by four users. The noise power δ^2 spectral density is set to -174 dBm/Hz. The shadow fading of large scale $\zeta_{j,i}^{e,t}$ is set to 8 dB. The ESs with the computation capacity of $F_j^{edg} = [3, 10]$ Gcycles/s are deployed together in a central location. The electronic devices that have the same computation capacity $f_i^{loc} = 0.5$ Gcycles/s are randomly distributed around the base station. We assume that the computation task data size D_i (in KB) obeys uniform distribution between $(250, 700)$ and the number of CPU cycles required by the electronic device and ES to calculate 1 bit data is $c_i = 30$ cycles/bit. The path loss factor $\varkappa$ is set to 3. The electronic devices estimate the amount of harvested energy based on the EH model given in Eq. 18, where the energy conversion efficiency $\nu = 0.6$, the energy transmission intensity $\rho = 0.6$ W, the antenna combination gain $G = 6$ dB, and the set of possible values of the distance between electronic devices and WET-AP is set to $\mathcal{G} \triangleq \{10, 12, 14, 16, 18, 20\}(m)$.

TABLE II
Scenario Parameters

Parameters	Value	Parameters	Value
δ^2	-174 dBm/Hz	p_i^{idel}	0.1 W
V_i^{max}	[10, 20] mJ	ϕ	3
$D_i(t)$	[250, 700] KB	ν	0.6
c_i	30 cycles/bit	ρ	0.6 W
f_{ij}^{loc}	0.5 cycles/bit	∂	2
F_j^{edg}	[3, 10] Gcycles/s	G	6 dB
$\mathcal{S}$	10^{-27}	ω_1	10
ω_2	0.5 units/ms	ε	6
ω_3, ω_6	0.8 units/KB	λ	0.001
ω_4, ω_5	0.9 units/KB	γ	0.99
τ	0.005	$\zeta_{j,i}^{e,t}$	8 dB
D	100000	λ_φ	0.001
λ_Q	0.002	ϵ_{off}	10^{-29}

In the learning algorithm, α is set to 0.2, the learning rate λ is set to 0.001, the discount factor γ of the reward function is set to 0.99 and the update factor τ of the action selection is set to 0.005. The Adam optimiser learning rate λ_Q for Q-value function network is 0.002 and the Adam optimiser learning rate λ_φ for policy network is 0.001. The main simulation parameters are listed in Tab. II.

B. Simulation Results

To simulate practical task generation patterns, three statistical distributions: Normal, Uniform, and Pareto are adopted for task size modeling. This setup enables a comprehensive assessment of the proposed algorithm's performance, with the results summarized in Fig. 3 to Fig. 4 for different offloading approaches in the MEC-assisted IoT system.

1) Convergence analysis of algorithms

Fig. 3 illustrates the convergence of the reward curves for the ECOP, CMDP, and RL algorithms during the training process. In the early stages of training, all three algorithms exhibit significant fluctuations in their reward curves. This phenomenon may be attributed to the randomness and exploratory nature of the initial policy parameters, which prevent the agent from sufficiently exploring the policy space and thus from stably learning an effective policy. However, as training proceeds, the reward curves gradually stabilize in later training episodes, accompanied by a progressive increase in reward values. This indicates that all three algorithms are capable of learning superior policies over time and achieving higher rewards in the environment. Such improvement likely arises from the agent's accumulation of experiential data, enabling it to extract more effective policies. Concurrently, the updates of policy parameters progressively converge to a near-optimal state, allowing the agent to execute actions more stably and attain higher returns. In summary, although the ECOP algorithm may exhibit oscillations during the early training phase, it gradually learns effective policies as training

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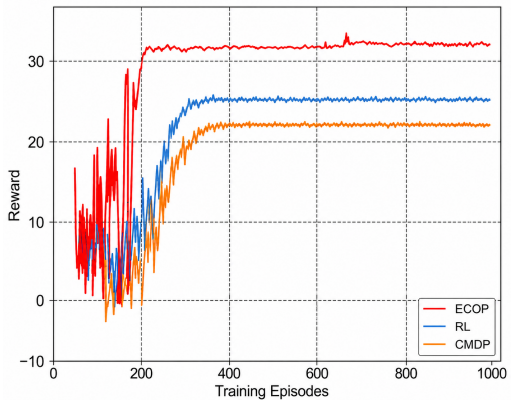


Fig. 3. Training curve overview.

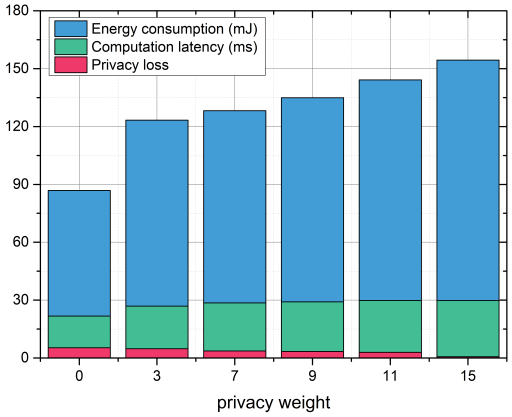


Fig. 4. Effect of different privacy weight.

advances and demonstrates favorable convergence and performance enhancement in later stages. These results underscore the effectiveness and robustness of the ECOP algorithm in addressing reinforcement learning problems.

2) The impact of different privacy weight ϑ sizes on algorithm performance.

As shown in the optimized objective function (Eq. 18), the privacy weight reflects the level of concern for location privacy protection by the electronic devices. This weight value depends on the subjective awareness of the user. To investigate the impact of different weight values on system performance: privacy loss, computation latency, and energy consumption, this section sets three different privacy levels: [0, 5] for low-level privacy protection, [5, 10] for mid-level privacy protection, and [10, 15] for high-level privacy protection. Six different values are selected for experimental verification, i.e., $\vartheta = \{0, 3, 7, 9, 11, 15\}$. Here, $\vartheta = 0$ indicates that all tasks generated in the current time slot will be computed locally, while $\vartheta = 15$ means that all tasks generated will be offloaded to the nearest ES for computation. Generally speaking, we set the task size to 500 KB and it follows a uniform distribution.

As illustrated in the experimental results (Fig. 4), as the weight value ϑ gradually increases, the privacy loss of the ECOP algorithm decreases. However, both latency and energy consumption increase. This phenomenon primarily occurs because the value of ϑ reflects the emphasis placed on location privacy protection. A higher ϑ signifies a greater focus on privacy protection by the electronic devices. Consequently, when executing tasks generated by such devices, the ECOP algorithm tends to process these tasks locally or minimize the number of ESs accessed by the device. Task queuing commonly occurs due to the constrained resources available on local devices, causing higher latency and greater energy consumption. To further investigate the influence of task size on performance, we conduct experiments with $\vartheta = 7$.

3) The impact of task size on algorithm performance.

A consistent rise in computation latency and energy consumption is observed across all algorithms as the volume of randomly generated tasks increases in the MEC-assisted IoT system. This pattern is visually demonstrated in Fig. 5 and Fig. 6. In addition, with the growing volume of computation tasks, the advantages of the proposed ECOP algorithm become more pronounced. Compared to CMDP and RL methods, it exhibits the lowest increase in both computational delay and power consumption. For example, when the task size obeys a Normal distribution and task size is 500 KB, compared with CMDP and RL algorithms, the ECOP algorithm can reduce the computation latency by 15.69% and 12.15% respectively. And it can reduce the task execution energy consumption by 9.28% and 6.05% respectively. Regardless of the distribution of task size, the ECOP algorithm demonstrates superior performance in computation latency and energy consumption compared to CMDP and RL algorithms.

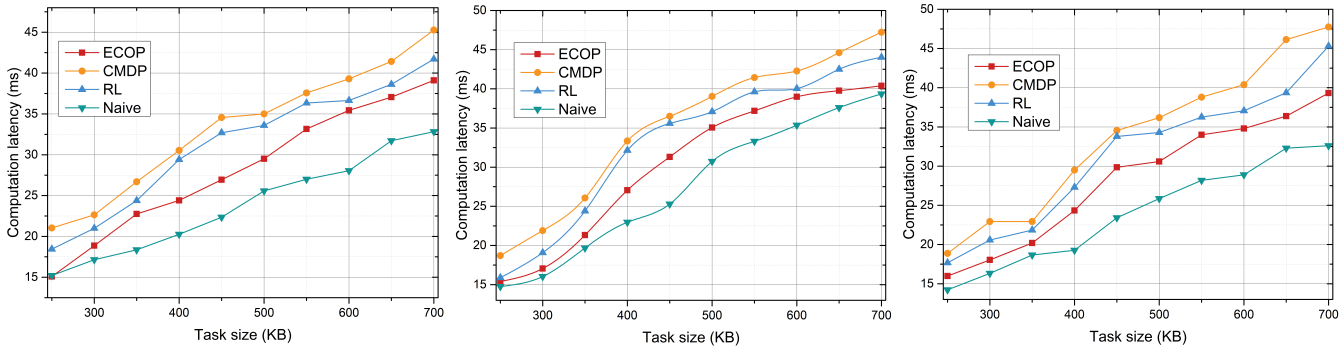
It is worth noting that the Naive offloading scheme does not consider the location privacy, so in comparison with the other three schemes, it achieves lower computation latency and reduced energy consumption. For example, when the task size obeys a Normal distribution and task size is 500 KB, the computation latency of CMDP, RL and ECOP are increased by 36.88%, 31.36% and 15.41% respectively compared with the Naive algorithm. Meanwhile, when the task size obeys a Normal distribution and task size is 500 KB, the energy consumption of CMDP, RL and ECOP are increased by 12.95%, 9.08% and 2.48% respectively compared with the Naive algorithm. The Naive scheme provides improved computation latency and energy consumption relative to the other three approaches, but this improvement is accompanied by severe privacy concerns.

Fig. 7 illustrates how the task size affects the task abandonment rates of various schemes. As the computation task size grows, so does the task abandonment rate. For instance, when the task size obeys a Normal distribution and task size changes from 400 KB to 500 KB, the task abandonment rate of the ECOP scheme increases by 50.00%. Compared with the proposed ECOP scheme, CMDP and RL schemes have a higher task abandonment rate especially when the task size becomes larger. It can be attributed to the use of RFEH technology in our scheme, in contrast to the adoption of green EH technology in others, which is associated with difficulties in ensuring energy supply quality and reliability.

As shown in Fig. 8, as the amounts of computation task continues to increase the privacy entropy under all algorithms increases approximately linearly. When the computation task

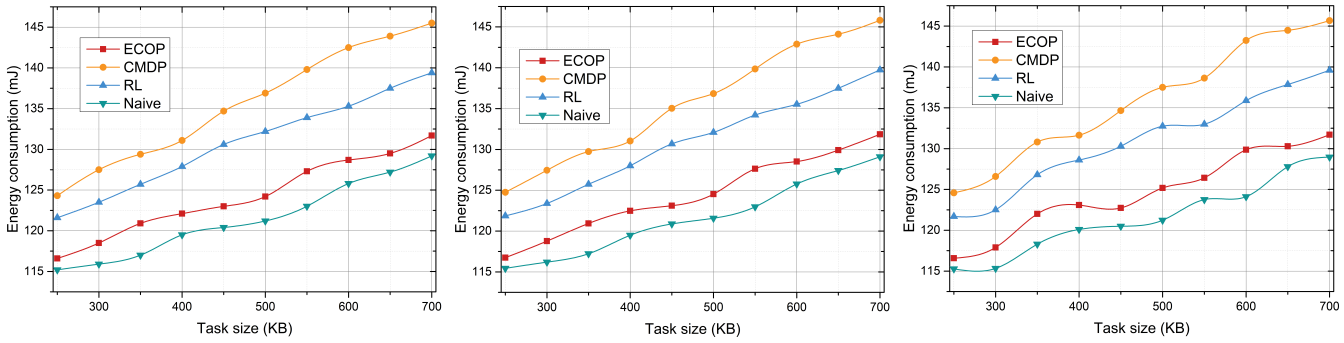
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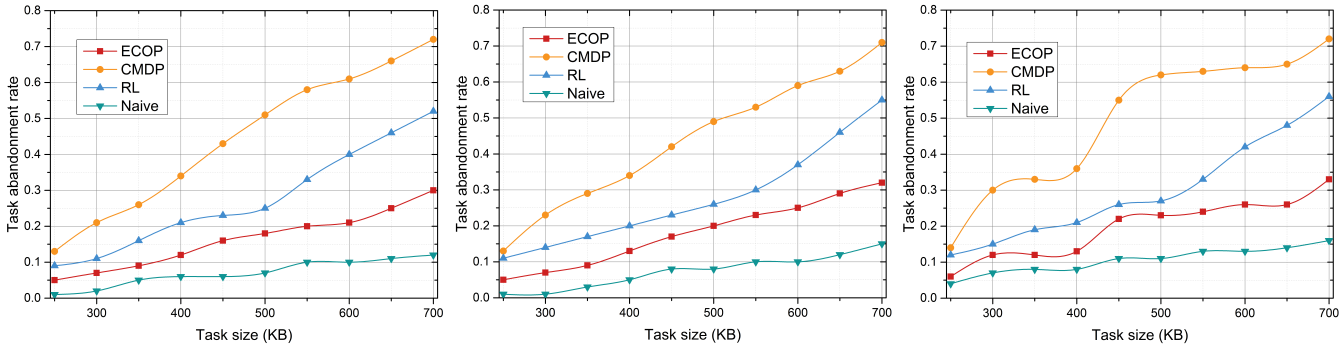
(a). Task size: Normal distribution (b). Task size: Uniform distribution (c). Task size: Pareto distribution

Fig. 5. Computation latency under varying task sizes.



(a). Task size: Normal distribution (b). Task size: Uniform distribution (c). Task size: Pareto distribution

Fig. 6. Energy consumption under varying task sizes.



(a). Task size: Normal distribution (b). Task size: Uniform distribution (c). Task size: Pareto distribution

Fig. 7. Task abandonment rate under varying task sizes.

volume is 500KB, the ECOP scheme we propose increases the privacy tiers by 19.656% and 14.82% respectively compared with CMDP and RL schemes. This is because our scheme comprehensively considering the impact of ES and WET-AP on location privacy leakage, while the CMDP and RL schemes only consider the impact of ES. The proposed ECOP algorithm has a greater improvement in improving system privacy rating.

As the computation task size increases, the system utility of the MEC-assisted IoT syatem grows approximately linearly across all algorithms. This trend is illustrated in Fig. 9, which shows how utility varies with task size. For instance, when the task size obeys a Normal distribution and task size is 500KB, the system utility of CMDP and RL schemes are increased by 24.17% and 8.59% respectively compared with the ECOP scheme. The proposed ECOP algorithm has a great improvement in improving the system utility.

4) The performance comparison of the ECOP and ECOP-No.

To provide a more comprehensive evaluation of the proposed privacy awareness offloading strategy, we compare it with ECOP-No, an algorithm that does not account for WET-AP-induced location privacy leakage. The performance differences are visualized in Fig. 10. From the simulation result, we can see that although the ECOP algorithm is somewhat higher than the ECOP-No algorithm with respect to computation latency and energy consumption, the ECOP algorithm can obtain a higher level of location privacy protection which is almost twice as much as the ECOP-No algorithm. Moreover, although the ECOP algorithm sacrifices a certain amounts of

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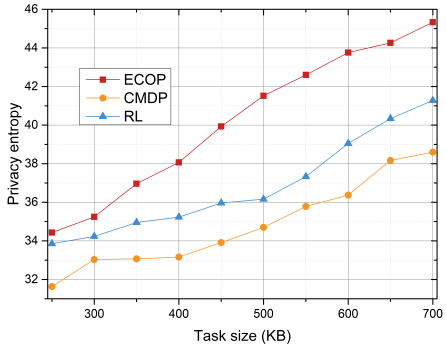


Fig. 8. Privacy entropy under varying task sizes.

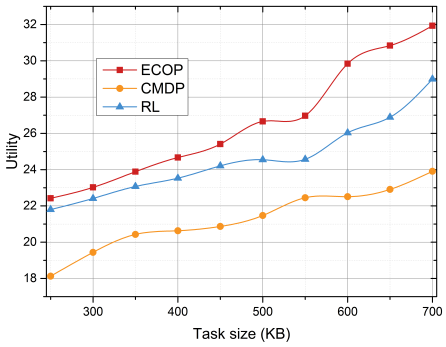


Fig. 9. System utility under varying task sizes.

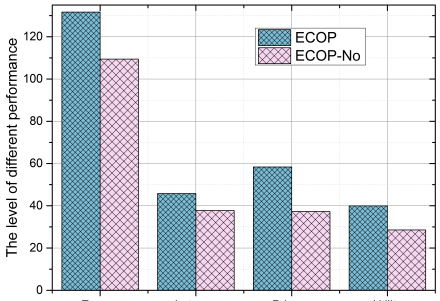


Fig. 10. Performance comparison of ECOP and ECOP-No.

computation latency and energy consumption, it has a higher system utility.

5) Performance analysis for failure rate

Task failure rate serves as a core evaluation metric for assessing the performance of task allocation strategies in MEC systems. Task execution failures primarily arise from two key factors: excessive resource utilization of MEC servers and deteriorated network conditions, both of which hinder servers from receiving and processing tasks normally. In addition, the hardware and parameter configurations of user electronic devices can also trigger task execution failures. This paper therefore investigates and verifies the task failure rate induced by insufficient computing and communication network resources. Fig. 11 demonstrates that the average task failure rate presents a gradual upward trend with the growth of task scales. The proposed algorithm ECOP achieves a lower task failure rate and superior performance compared with the baseline algorithms, including CMDP, RL, and Naive.

Generally, all comparative algorithms require an exploration stage to optimize offloading decision-making, and improper decisions generated during this stage inevitably cause partial task failures. This phenomenon becomes more prominent as the number of tasks increases: system overload occurs under heavy task loads, and minor decision errors will be further amplified to deteriorate system performance. In contrast, the proposed algorithm ECOP can dynamically allocate reasonable computing and communication resources adapting to diverse network scenarios, which effectively reduces the task failure rate and enhances system stability.

In summary, the above simulation results show that the energy-efficient and privacy awareness task offloading scheme can effectively reduce the privacy leakage of the MEC-assisted IoT system while achieving lower system energy consumption and computation latency.

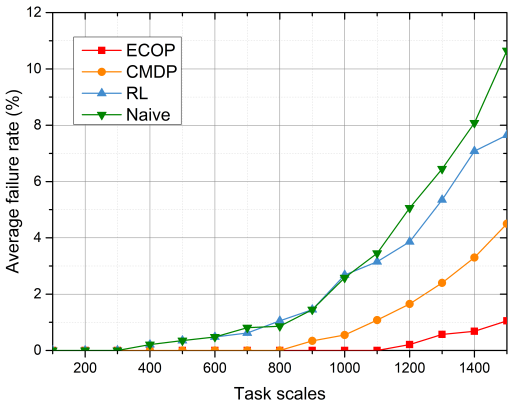


Fig. 11. Effect of number of tasks on task failure rate.

C. Discussion

Analysis of signaling overhead. Our framework adopts a centralized training and decision-making paradigm at the MEC server. Only lightweight real-time state information (i.e., battery levels, channel gains, and task arrival status) is uploaded by IoT devices at each time slot. The DRL model parameters are updated offline at the MEC side and do not require frequent transmission between the server and devices, thereby incurring negligible signaling overhead for IoT terminals.

Analysis of model update frequency. The SAC model is updated periodically in the offline training phase rather than in every time slot. During online deployment, the well-trained model generates offloading decisions in a feed-forward manner without parameter iteration. Such an episodic update strategy effectively reduces computational and update overhead and avoids frequent model retraining in dynamic environments.

Analysis of real-time feasibility. The online inference of the SAC policy only involves lightweight matrix computation. Once the model is fully trained, the MEC server can output optimal joint offloading and resource allocation decisions within milliseconds. This ultra-low online latency satisfies the real-time requirement of practical MEC-enabled IoT systems. In addition, the device-side only executes received control commands without additional training or inference burden, further ensuring the practicability of the proposed framework.

VI. Summary and Future Work

This paper comprehensively addresses the challenges of computation latency, energy consumption, and location privacy in MEC-assisted IoT systems by formulating an energy-efficient and privacy awareness task offloading optimization problem. To solve this optimization objective function, we transform the problem into an MDP problem, and also design a mobile computing offloading scheme with privacy protection

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by combining DRL and RFEH techniques. In this scheme, the offloading strategy of electronic devices depends solely on the current state which specifically includes the assignment of subchannels, the battery level, the amounts of harvested RF energy and the generated computation tasks. The task offloading rate is optimized by observing the state of the MEC-assisted IoT system in each time slot, which improves the computing performance of the system. Moreover, the location privacy is better protected by driving the offloading frequency deviate more from the original offloading frequency. Additionally, we considered the impact of RF-enable WET-AP on location privacy leakage for the first time. Simulation results demonstrate that the proposed algorithm effectively balances the trade-offs between privacy protection, computational latency, and energy consumption, offering significant improvements over existing approaches. However, numerous challenges remain for future research, including accounting for the impact of mobile states of electronic devices on network topology, establishing a task-dependency model for non-independent tasks, introducing task migration to achieve network load balancing against abnormal network traffic, and developing generalized optimization functions to accommodate diverse requirements.

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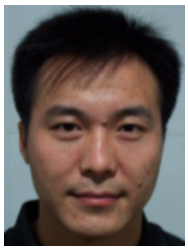
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